Dylan L Randle

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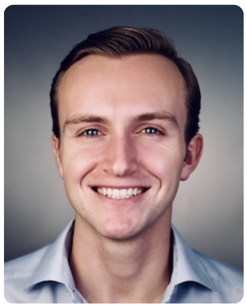

Summary

Technical lead with 8+ years of experience in AI, robotics, and machine learning. Expert in developing closed-loop policies for dexterous manipulation using imitation and reinforcement learning. Proven track record building and deploying AI/ML systems for robotics, computer vision, and natural language processing.

Experience

Amazon Robotics — North Reading, MA, USA

Senior Applied Scientist — *Apr 2024–Present*

- Tech lead for closed-loop imitation and reinforcement learning systems for dexterous bimanual manipulation.
- Delivered Amazon's first bimanual manipulation policies, achieving a 92% success rate on unseen items. The technology was demonstrated to Jeff Bezos, Andy Jassy, and the Board of Directors.

Senior Data Scientist — *Apr 2023–Apr 2024*

- Developed production ML systems for robotic manipulation, including grasp generation, damage prediction, and box packing.
- Deployed the first learned action model for Amazon's Sparrow robot, delivering performance improvements of 35% and savings of $10 million per year.

Data Scientist II — *Jul 2020–Apr 2023*

- Developed an ML path-planning optimization system for large-scale mobile robot fleets.
- Published research demonstrating a 15% throughput increase and identifying $150 million in annualized efficiency gains for mobile robot fleets.

Data Scientist I — *Jun 2019–Aug 2019*

- Developed an AutoML system for training, evaluating, and interpreting models trained on trillion-row robotics datasets.
- The system was used by multiple scientists to accelerate research and analysis workflows.

Hubdoc — Toronto, ON, Canada

Data Scientist — *Feb 2017–Jul 2018*

- Started and led the ML team from ideation through the company's $70 million USD acquisition.
- Developed an ML-based NLP system for automated data extraction from unstructured financial documents.
- Reduced data-extraction time from hours to seconds.

BMO Capital Markets — Toronto, ON, Canada

Financial Products Analyst — *Jun 2014–Aug 2014*

- Developed an interest-rate swap and swaption delta-hedging optimization algorithm.
- Uncovered market opportunities for fixed-income traders.

Education

Harvard University — Cambridge, MA, USA

Master of Science in Data Science · GPA 4.0 — *Aug 2018–May 2020*

- Thesis: Unsupervised Neural Network Methods for Solving Differential Equations.
- Recognized with Scholarship in Applied Computation and Distinction in Teaching.
- Research and coursework focused on machine learning.

Education (continued)

University of California, Berkeley Berkeley, CA, USA
Bachelor of Science in Industrial Engineering & Operations Research · GPA 3.9 *Aug 2012–May 2016*

- Graduated with High Honors (magna cum laude) and received the Frank Kraft Award.
- Inducted into Phi Beta Kappa, Tau Beta Pi, and Alpha Pi Mu.
- Coursework focused on statistics and optimization.

Publications

- **Demonstrating Multi-Suction Item Picking at Scale via Multi-Modal Learning of Pick Success** C. Wang, J. van Baar, C. Mitash, S. Li, D. Randle, W. Wang, S. Sontakke, K. E. Bekris, K. Katyal. RSS 2025.
- **MuST: Multi-Head Skill Transformer for Long-Horizon Dexterous Manipulation with Skill Progress** K. Gao, F. Wang, E. Aduh, D. Randle, J. Shi. ICRA 2025.
- **Learning Object Properties Using Robot Proprioception via Differentiable Robot-Object Interaction** P. Y. Chen, C. Liu, P. Ma, J. Eastman, D. Rus, D. Randle, Y. Ivanov, W. Matusik. ICRA 2025.
- **Avoiding Object Damage in Robotic Manipulation** E. Aduh, F. Wang, D. Randle, K. Wang, P. Shah, C. Mitash, M. Nambi. IROS 2024.
- **DEQGAN: Learning the Loss Function for PINNs with Generative Adversarial Networks** B. Bullwinkel*, D. Randle*, P. Protopapas, D. Sondak. ICML 2022, AI for Science · Equal contribution.
- **Unsupervised Learning of Solutions to Differential Equations with Generative Adversarial Networks** D. Randle, P. Protopapas, D. Sondak. arXiv:2007.11133 · 2020.
- **Unsupervised Neural Network Methods for Solving Differential Equations** D. Randle. Master's thesis, Harvard University · 2020.

Projects

- **Rubik's Cube Solving Robot** *(Nov 2024)*: Built a robot that solves a cube in under three seconds using camera-based perception, Kociemba's planning algorithm, and custom stepper-motor hardware.
- **Real-Time Quantitative Trading System** *(Jun 2023)*: Built and deployed a live quantitative trading system that uses real-time NLP analysis of news articles to generate market signals and execute trades.
- **Golf Swing Analysis with Computer Vision** *(Jul 2020)*: Applied monocular 3D human-pose estimation to analyze body mechanics throughout a golf swing.
- **ResNet Variational Autoencoder** *(Jun 2020)*: Trained a residual-network VAE on CelebA to learn a generative latent representation and synthesize faces.
- **Differentiable Neural Architecture Search** *(Dec 2019)*: Partnered with Google AI to adapt and evaluate DARTS across materials science, astronomy, and medical-imaging datasets.
- **Interpretable Reinforcement Learning** *(Dec 2019)*: Distilled black-box policies into readable decision sets using DAgger and quantified the interpretability-performance tradeoff in an HIV treatment simulator.
- **Distributed YouTube-8M Classification** *(May 2019)*: Trained bidirectional LSTM video classifiers with Spark, TensorFlow, Elephas, HDFS, and a custom AWS EMR cluster.
- **Causal LSTM Microbiome Modeling** *(May 2019)*: Used sparse LSTMs, neural Granger-causality analysis, and Bayesian optimization to compare microbial interactions in healthy and IBD mice.
- **Bayesian Generative Adversarial Networks** *(Dec 2018)*: Reproduced and evaluated Bayesian GANs using stochastic-gradient HMC, demonstrating mode diversity and semi-supervised learning.
- **Twitter Troll Detection** *(Dec 2018)*: Built NLP classifiers for identifying Internet Research Agency tweets, achieving 95.9% accuracy on a temporally separated test set.
- **Automatic Differentiation from Scratch** *(Dec 2018)*: Built and published the dragongrad Python package with forward- and reverse-mode differentiation, Jacobians, and gradient-based optimization.

Technical Skills

- **Languages:** Python, C++, JavaScript/TypeScript, SQL.
- **Libraries:** PyTorch, Keras/TensorFlow, OpenCV, Open3D, Pandas, NumPy, SciPy, scikit-learn, React.
- **Platforms:** AWS, Docker, Firebase, Linux, macOS.